

Module 1

Fracture problem

- Plate geometry
- Notch, aggregates
- Material properties
- Boundary conditions
- Loading conditions
- Phase-field model
- Mesh refinement

Module 2A

Dataset generation

- Vary notch height
- Vary aggregates
- Solve load steps
- Displacement field
- Damage field
- Save every step

Module 2B

Pre-processing

- Extract nodal data
- Assemble archive
- ML-ready dataset

Module 3

Develop ML model

- Define the task
- Input/output spaces
- Model architecture
- Train on the data
- Validate and test
- Fast prediction